

StaffControllerTest.java — Tanglish Explanation

■ Overview:

Ithu `StaffControllerTest` class — unga `StaffController` endpoints test panna use panra JUnit test file da. Spring Boot la `MockMvc` use panni controller ah test panrom, database la actual data affect aagama.

■ Important Annotations:

@SpringBootTest – entire Spring context load pannum.

@AutoConfigureMockMvc – MockMvc auto-configure agum for controller testing.

@Transactional – Test mudinja udane database la rollback aagum (data safe!).

■ Autowired Components:

- **MockMvc**: controller endpoints test panna.
- **ObjectMapper**: Java object → JSON convert panna.
- **StaffRepository**: DB la existing staff records check panna.

■ Test Cases Explanation:

1■■ ****testGetAllStaff()****

All staff records get panna `/api/staff` endpoint test pannum. Expected result → status 200 OK + JSON content.

2■■ ****testGetStaffById()****

Existing staff ID eduthu, `/api/staff/{id}` call pannum. ID irundha status 200 OK, illa na console la warning print pannum.

3■■ ****testAddStaff()****

New staff object create pannum (`JUnit Staff`), adha `/api/staff` ku POST pannum. Expect pannradhu: 201 Created + name match.

4■■ ****testUpdateStaff()****

Existing staff oda designation change pannum (`Senior Manager`) via PUT request. Expect: 200 OK + updated value verify.

5■■ ****testDeleteStaff()****

DB la oru staff irundha adha `/api/staff/{id}` DELETE pannum. Expect: 204 No Content (delete success).

■ Summary (Tanglish):

Ithu unit testing file da. Data DB la permanent a store aagama rollback aagum. Every CRUD operation test pannum — GET, POST, PUT, DELETE ellam proper-a validate pannum.

■ Simple Meaning:

StaffController correct-a work panratha check panna ithu main tool da. Real data change aagama complete test panna safe setup.